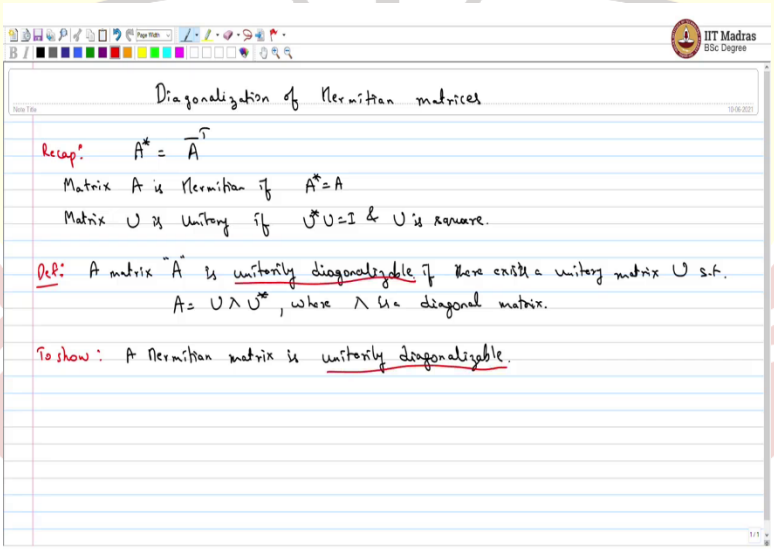
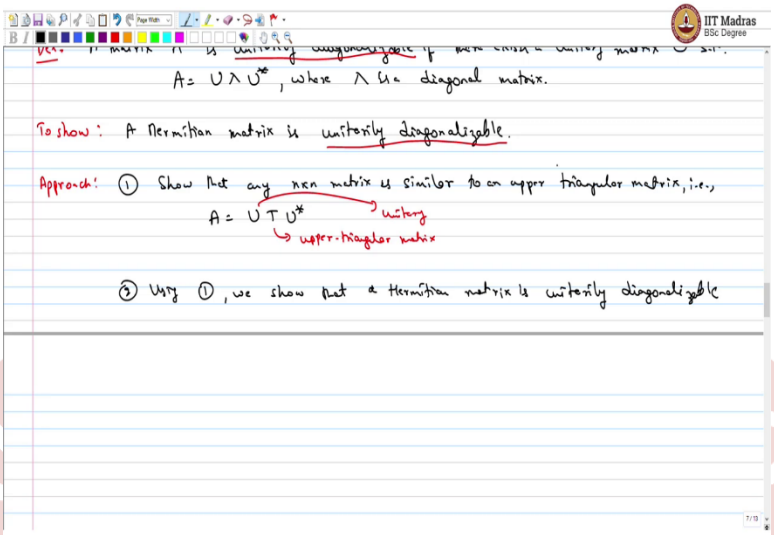
Machine Learning Foundations
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Diagonalization of Hermitian Matrices (Part 1)

Welcome to lecture 4 in week 5 of linear algebra segment of machine learning foundations course. We will *t the discussion of a very important topic in this lecture, which is Diagonalization of Hermitian Matrices. So, we will define what it means to say a matrix is unitarily diagonalizable and then show that the final step is to show that Hermitian matrix is unitarily diagonalizable.

And as a proof for the approach is to show that any matrix, not necessarily Hermitian, any n cross in complex matrix can be upper triangularized. We will define it concretely during the lecture. And I use this fact for any matrix, use this fact that any matrix is upper triangularizable to show that Hermitian matrix is diagonalizable. So, that is the approach we are going to adopt to show our mission to show Hermitian, to show diagonalizability of Hermitian matrices.

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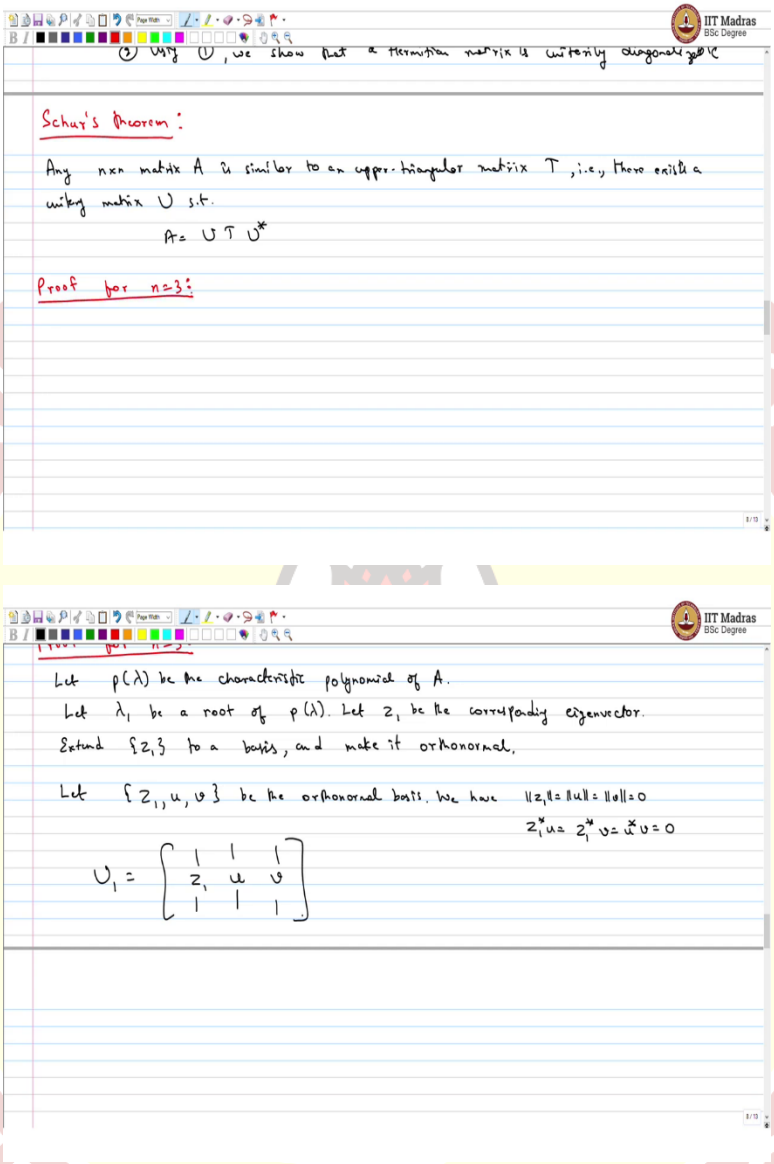


Just a quick recap, what is A *, A * is just a conjugate transpose and matrix A is Hermitian if A * = A, matrix U is unitary if U * U sign identity and U is square. A matrix A is unitarily diagonalizable if there exists a unitary matrix U such that A = U λ U *, where λ is diagonal matrix.

Now, what we want to show is a Hermitian matrix is unitarily diagonalizable. So, the approach is; first, we show that any n cross n matrix is similar to an upper triangular matrix. So, we are not talking about Hermitian matrices, this is a more general claim. That is A = U T U * where T is an upper triangular matrix, U is unitary. Second, using 1 we show that Hermitian matrix is unitarily diagonalizable. So, that is the approach.

So, we will prove the more general theorem, which shows that I can take any n cross n matrix and find a unitary matrix such that I can transform this matrix into upper triangular matrix. This is like saying that both A and T have the same Eigen values.

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So, the more general theorem is called Schur's theorem. This shows that any n cross n matrix A is similar to an upper triangular matrix T that is there exists a unitary matrix U such that A = U T U *. Now, we will prove this for a simple case of three cross three matrix, but ideas are so general in this case that it is fairly straightforward to extend this to the more general case.

Proof for n = 3. So, what is coming next is we will prove this first and then do an example where we go through the procedure adopted in the proof to come up with this triangular matrix. And then later we will show that Hermitian matrices are unitary diagonalizable. Given Schur's theorem, it is fairly straightforward to show that Hermitian matrix is unitarily diagonalizable.

You just have to apply Schur's theorem to A and A * and then use the fact that A * = A to conclude what the main claim, to conclude diagonalizability, whereas upper triangular is, to show that any matrix is similar to upper triangular matrix which is this Schur's theorem requires a lot more work.

So, let us understand, I mean, how the proof proceeds. And this proof is important simply because it also gives a procedure to find an upper triangular matrix for any given matrix such that, I mean, for any given matrix A, it gives the procedure to follow or an algorithm to follow so that I get a triangular matrix such that A and T are similar.

So, let us begin the proof. Let p of λ be the characteristic polynomial of A. So, obviously, what are the roots of this equation, they are the Eigen values. Let λ 1 be a root, just pick one of the roots, p of λ. Let Z1 be the corresponding Eigenvector. Now, I can always take a vector and extend it to a basis. So, what you do is, extend Z1 to a basis and I can also make it orthonormal using the Gram–Schmidt procedure. You are familiar with this from an earlier course in this degree program. So, let us make it orthonormal.

So, let, for simplicity, let us just call Z1, let us assume Z1 was picked such that it is already unit length. So, Z1 u, v be the orthonormal basis. In particular, you have Z1 transpose, not transpose = 0 and so on. I mean, if I pick anything from this basic, so I will not write this. So, if I pick any two vectors from this basis, they are orthogonal. And the length of each vector is 1. So, I can write it out, we have Z1 * u = Z1 v = 0. So, they are orthogonal and of unit length. Now, let me *t with the candidate unitary matrix. I will put Z1, u and v as columns.

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So, let me call this. And now, what happens when I act A on this. What is AU1? AU1 is A applied to Z1, u, v which is = to AZ1. What is AZ1? Z1 is an Eigenvalue corresponding to λ 1. So, this is what we have. So, here, since AZ1 = λ 1 Z1, so that is what we have.

Now, what happens to this product? Consider this product, U1 * A U1. And what is U1 *? I have to transpose and then take conjugate. So, what is the transpose? Z1 will become a row, u, v and then I take. Now, this into AU1 from the above is what, I have λ 1 Z1, Au, Av. Now, I want to claim that this is = to λ 1, then 0s. This is the most important, but this is something with a matrix here. Let us understand these components. This some entries that are not necessarily 0.

However, what we got is, the most important bit is, we got all entries except the very first entry in column 1 became 0. Now, why is this true? Why? We will answer this question why. So, for this you will have to look at this product. So, Z1 bar transpose into λ 1 Z1 the answer lies here. Z1 bar transpose λ 1 Z1 = λ 1 into Z1 bar transpose Z1, but Z1 is unit length so this is = to λ 1. So, this is the first bit.

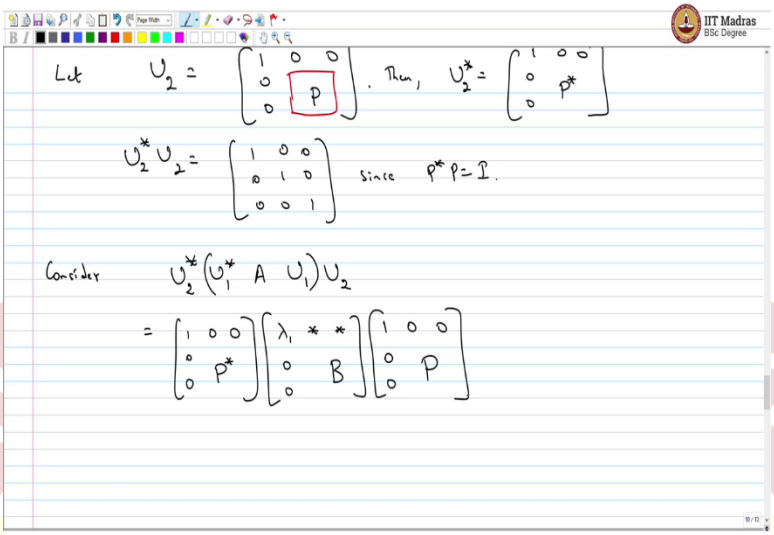
The second bit is, u bar transpose Z1 = 0 and v bar transpose Z1 = 0. This just means that the very first column, the first entry becomes λ 1, the rest of the two entries here becomes 0. So, this is coming from here, and this is coming from here. So, we have got, in general, n minus 1 cross n minus 1, but in this particular case, it is 2 cross 2. You could have *ted with the n cross n matrix and got to n minus 1 cross n minus 1, just doing it for 3 cross 3. What we have is, so U1 * AU1 =, you got this and some entries here and a matrix B.

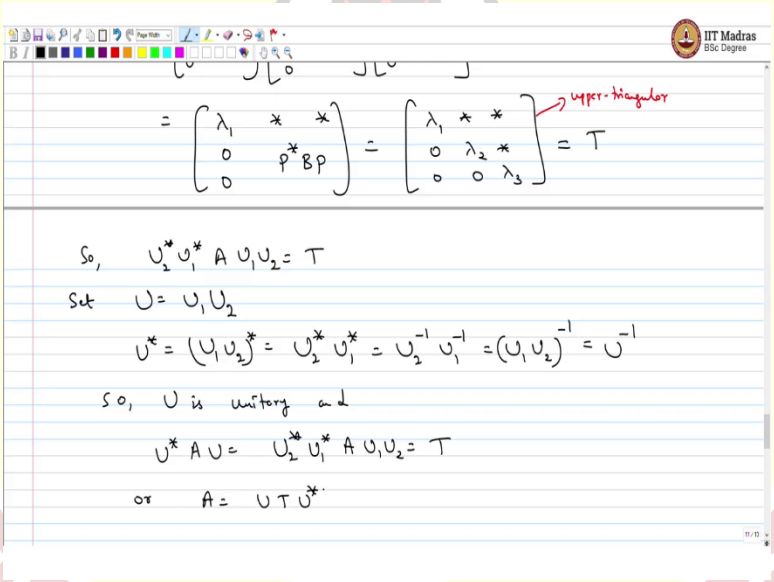
Now, repeat the procedure with B, just take the smaller dimensional matrix B, to get an Eigen value λ 2 of B and unitary matrix say P such that, what you would get is, see instead of U1 * AU1 the corresponding matrix B * BP this will be = to λ 2 and some number we have got to 1 cross 1 now, instead of B before I just have 1 cross 1 which is just some λ 3 and this will be some number, * denotes any number and then this is 0.

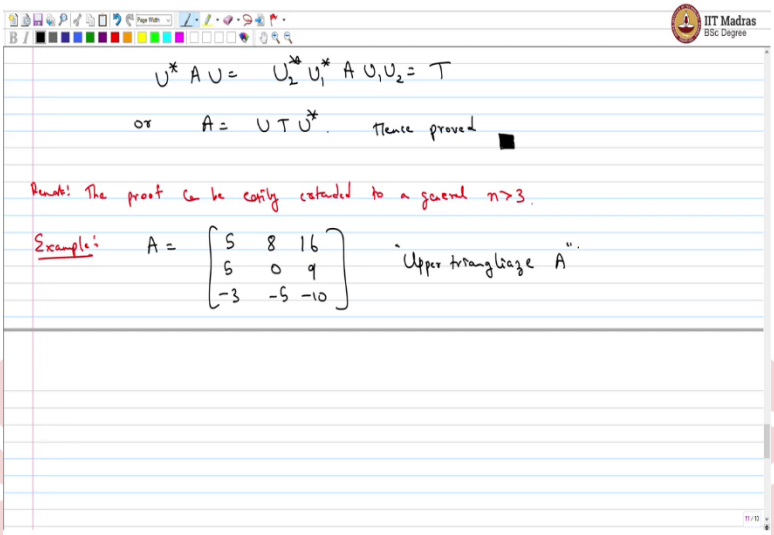
We follow the same procedure. What happened if you recall in the earlier procedure is when I do U1 * AU1 I got a column, the very first column, I got a non-zero entry in the first column which is an Eigen value, not non-zero, but I got the Eigen value in the first entry and then I got 0s in the remaining part of the column.

And so, similarly, I do it for B, same procedure, I get an Eigen value λ 2. I extend it, I mean, I take that eigenvector corresponding to λ 2, extend it to a basis and do the exact same procedure put some eigenvectors, put some basis vectors in a column matrix and call it P * and so on. So, when you repeat this procedure, what I get is P * BP = this.

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Now, what I am going to do is, let, I am going to make up another matrix U2, which is like this, which is going to, so the point is we are working with the matrix A. So, we will have to relate this matrix P that we got to a bigger size. So, let us do it like this 0, 0, 0, 0 and I will put P * here. So, the 2 cross 2 matrix P * sets here. Now, what is U2 *? Then U2 * = 1, 0, 0, 0, 0 and then, I have made a mistake, so this is U2, the P, now U2 * will have a P *.

Now, what is U2 * U2 is you can directly check. P is a unitary matrix. So, this is just since P * P = identity matrix. Now, we are nearly done with the claim. So, now, what we have to show is now consider this product, I have two matrices. I made up a matrix U1 and then I made up a matrix U2.

So, I am going to take U2 *, U1 * and multiply with A and then U1, U2. Look at this product. We already know U1 * AU1 from the first step, U1 * AU1 has this particular form. So, this is the same as U2 * is having this form 1, 0, 0, 0, 0 and a P * here. U1 * AU1 from the earlier discussion is like λ 1 0, 0, you have this matrix B here and some entries here. And then U2 is 1, 0, 0, 0, 0, P.

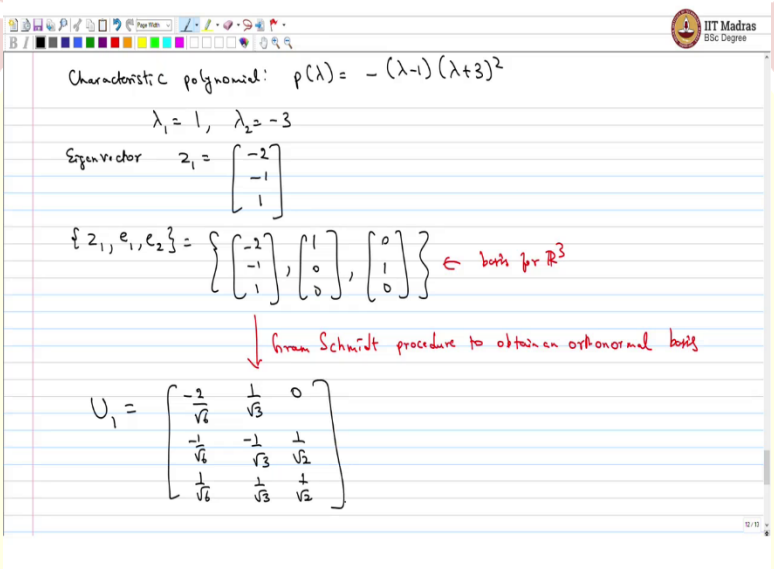
Now, this product just becomes λ 1 some numbers, this is 0 and then here you get P * BP. But this form of P * BP is something that we recovered here λ 2, λ 3 along the diagonal. So, it is also upper triangular. So, what we have done is, incrementally, we have arrived at a matrix like this and some entries here and this is the upper triangular matrix. So, what we have concluded is, so U2 * U1 * AU1U2 is an upper triangular matrix.

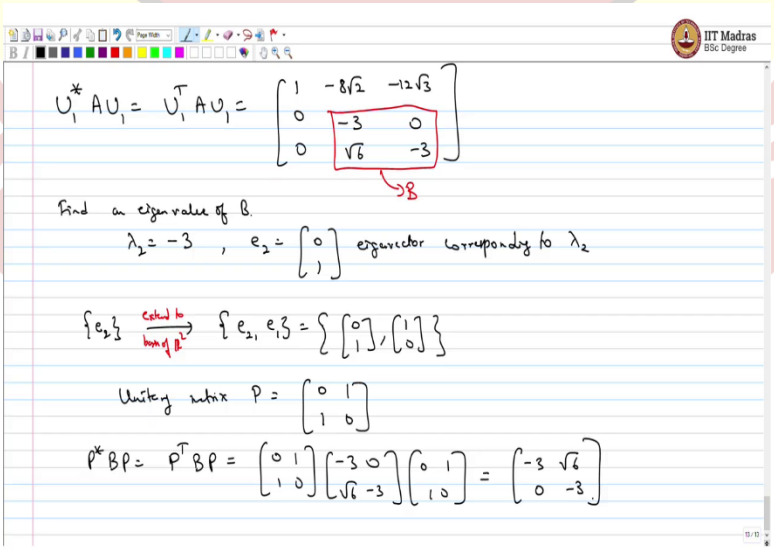
Are we done? There is one final detail to work out. So, what is the final U? So, set U = U1 into U2, U * = U1U2 * = U2 * U1 * = to U2 inverse U1 inverse = to U1U2 inverse is = to U inverse.

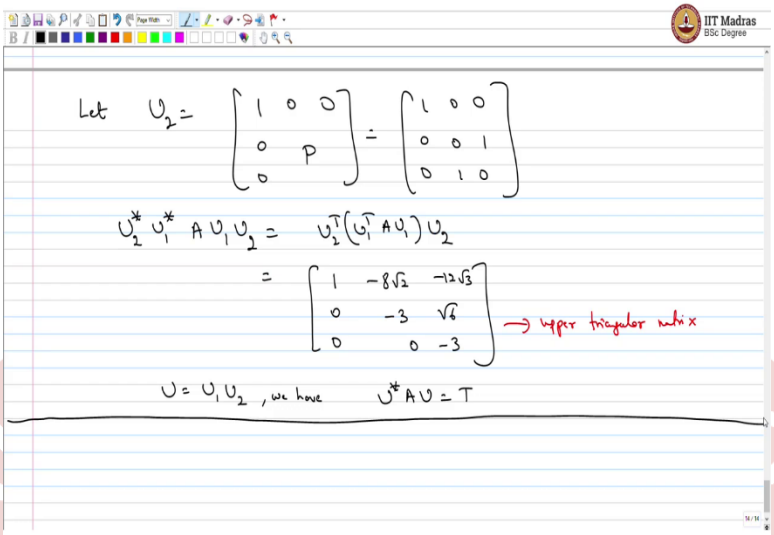
So, U is unitary and U * AU = U2 * U1 * AU1U2 is upper triangular or A = U T U *. That is the end of the proof.

So, that is Schur's theorem, but the proof is fairly general. I mean, remark, the proof can be easily extended to a general n greater than 2. It is very, very straightforward. I think it can be easy homework. What we will look at next is an example. Let us look at an example where this procedure comes into actions. Let us put this matrix 5, 8, 16, 5, 0, 9, minus 3, minus 5, minus 10. Problem is to upper triangularize A, not just diagonalize, but just upper triangularize A.

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So, let us *t with the characteristic polynomial. So, this is just a rehash of the proof just going to go through the same steps. So, what is the characteristic polynomial? You can pause and work this out on your own. I am not going to give out all the details where it is fairly straightforward. Just check that this is the correct characteristic.

So, it is clear that you have a repeated Eigen value. So, it may not be diagonalizable. So, the Eigen values are λ 1 = 1, λ 2 = minus 3. And eigenvector, let us say Z1, you can check this. You will get it as minus 2, minus 1 and 1 10 by equating A minus I times x = 0. Now, if you recall the procedure in the proof, I take Z1 and add e1, e2, so this is the set minus 2, minus 1, 1 and I am going to just add 1, 0, 0, 0, 1, 0. Now, this is a basis for R3.

Now, what we will do is apply Gram–Schmidt procedure to obtain an orthonormal basis. And that, if you recall, the proof is my first matrix U1 that turns out to be, for the very first vector, you just have to normalize it. So, it is minus 2 by square root 6, minus 1 by square root 6, 1 by square root 6. But other parts require more work. So, you have to do Gram–Schmidt on your own. I am just going to throw answer 0, 1 by square root 2, 1 by square root 2. I mean, it is easy to check if this is a unitary matrix, just take the dot product of the columns and then you will know.

Now, what is U1 * AU1. There are no complex numbers here just U1 transpose AU1. And this turns out to be, if you see, by construction, this is 1, 0, 0. And then you will get some numbers here. The numbers turn out to be minus 8 square root 2 and minus 12 square root 3. But the important bit is you get a smaller matrix here.

Let us focus on this matrix next. So, the next process is, we will call this B. Now, find an Eigen value of B. Here, so you get λ 2 = minus 3, eigenvector corresponding to λ 2. Now, it is very easy to extend this to orthonormal basis. Take e2 and to make it an orthonormal basis all you have to do is e1 which is like take 0, 1 and add 1, 0 that is the easiest way. So, this extend to here basis of R2.

From this I make the unitary matrix. So, the unitary matrix U, not U here, P is going to be just these basis vectors. And now P * BP = P transpose BP and this turns out to be this matrix. So, we have P. So, how do we make up U2? So, let U2 = we put here, which is the same as 1, 0, 0, 0, 0, P is 0, 1 and 1, 0. This, you can recognize this as a permutation matrix if you are familiar with it. But e1 you need not, you do not have to worry.

But the important bit now is, let us look at U2 * U1 * AU1U2 this is the matrix to look at, which is U2 transpose U1 transpose AU1 into U2. This I will leave it as home work. It is simple matrix multiplication. It turns out to be this matrix, 12 square of 3, 0, 0 and you just got P * BP that comes here. You would not have to multiply. All of the ingredients are already here. U1 * AU1 is something that you know and then you just have to patch in P * BP. And this is the upper triangular matrix.

So, you have setting U = U1U2 we have U * AU = to an upper triangular matrix. So, that is upper triangularization for U. We went through this procedure to take, in this particular case, if you observe this matrix that we had is not realize, is a real matrix, but it is not symmetric. So, we did not get a diagonal matrix in that. We got an upper triangular matrix. In particularly, it had repeated Eigen values. So, if we had three distinct Eigen values, we would have got a diagonal matrix, but in this particular case, it had an Eigen value that got repeated and we did not get enough eigenvectors to diagonalize it fully. But, for this matrix, we could still upper triangularize it.

So, we will end this lecture with a recap of what we learned today in this video. So, what did we learn? We understood what it means to say when a matrix is unitarily diagnosable. The next thing was to, was a proof with also followed by an example of the fact that any n cross n complex matrix is upper triangularizabal. This is the well known Schur's theorem.

We actually proved this for the three dimensional case, but the proof is so general that it can be easily extended to any dimensions and, any higher dimensions and then we actually, the proof gave

a procedure for upper triangularizing a matrix. So, we have followed the steps that we adopted in the proof to arrive at the upper triangularized form of a matrix for 3 cross 3 matrix, a complex matrix that we made up.

Next, coming next is we are going to Schur's theorem to show that any Hermitian matrix is unitarily diagonalizable that is coming in the next lecture.

